

Dhruv Nigam

MACHINE LEARNING ENGINEER

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Work Experience

Dream 11

LEAD MACHINE LEARNING ENGINEER

Mumbai

July 2022 - Present

- Trained **200M-parameter cross-attention transformer (Lumos)** on **1.7T user tokens** from **250M users**. Replaced **50+ task-specific models**. **+3.15% DAU**, **-2.47% offer spend**. Built **Accio**, a Delta-native streaming data loader for distributed training on **8x A100s**. [\[tech report\]](#), [\[blog\]](#)
- Built **real-time AI voice agent** for live cricket streams — **speech-to-speech over WebRTC**, **diffusion-based avatar**, **sub-2s latency**. **1.5M min watchtime**, **1.1M concurrent viewers** at peak (IPL 2026). [\[blog\]](#)
- Fine-tuned **Llama 8B with GRPO** for a sports banter bot. Trained a **DeBERTa-v3 reward model** on **222K preference pairs** to score response engagingness. Served via **vLLM on A100** at **3,000 RPM**, **400ms TTFT**. **+16% feature retention**.
- Built **Pacman**, a **personalized promotion platform** for **200M users**: supervised model + policy layer + PM tooling. Allocated **\$90M+ annually**, zero-downtime system. **+14% WAU**, **85% ROI**.
- Developed **ForeCal**, a probability calibration technique, **published at KDD 2024** [\[paper\]](#). Improved accuracy of churn prediction model used for offer targeting. Reduced calibration error by **50%**, **0.4% revenue upside**.

Protium

DIRECTOR - DATA SCIENCE AND ENGINEERING

Mumbai

Feb. 2020 - June 2022

- 12th employee and **founding ML Engineer**. Built deep learning models on **5M+ user time series records**, beating Credit bureau scores by **12% AUC**. Built a **Bayesian income estimation model** that inferred borrower income from partial loan observations via MAP inference over latent tenure and rate parameters. Together, these models underwrote over **\$500M in loans across India** without human intervention.
- Built the **ML platform** that hosted 52+ models as FastAPI endpoints for **real-time feature engineering and inference** supporting up to **10,000 rpm**.

CLSA

QUANTITATIVE TRADER, INVESTMENT BANKING

Mumbai

Nov. 2017 - Feb. 2020

- Founding member of a 3-person ML team. Built **model-based trade execution strategies** handling **\$1 bn/day** in client trades at **sub-10ms latency**. Developed a **Java-based event-driven system** for real-time feature calculation and low-latency ML inference.
- Developed a profitable quantitative trading strategy (**Sharpe 1.3**) using **cointegration-based time series statistical arbitrage** in US futures. Traded over 2 years with **\$10M in capital**, **\$4M net profit**.

Open Source, Talks & Writing

Talks ForeCal at **KDD 2024** (Barcelona), LUMOS at **Swiggy** (invited), GPU Fundamentals at **MumbaiPy**, etc. [\[all talks\]](#)

Writing Articles on miltrenches.substack.com: Algorithmic Intensity, RL from Scratch, Voice Agent Lessons Learned.

Open Source Merged PRs in [LiveKit Agents](#) and [TRL](#) (HuggingFace).

Education

India Institute of Technology - Bombay (IITB)

Mumbai

BACHELORS AND MASTERS IN ELECTRICAL ENGINEERING, SPECIALIZING IN SIGNAL PROCESSING. THESIS: [Single Channel Blind Dereverberation of Speech Signals](#)

2011 - 2016

Technical Skills

ML & GenAI PyTorch, Hugging Face, vLLM, LangGraph, LiveKit Agents, Scikit-learn

Data & Infra Apache Spark, PySpark, Databricks, Delta Lake, Apache Kafka, Apache Flink, Pandas

Cloud & MLOps AWS (EMR, S3, Glue, Lambda), GCP (BigQuery, Cloud Run, GKE), Docker, Kubernetes, FastAPI, MLflow, Ray